

BUILD PLAN • SMALL BUSINESS AI APPLIANCE

Remote Control Build Plan

A box that runs a small business by operating the apps that business already uses — on a real Android handset with a real SIM — and a subscription that exists because those apps never stop changing.

BUYER	PRICE	SOLD	STACK
Small business owner	~\$200/mo + hardware	Subscription to upkeep	Fully open source
FIRST SHIP	Software, not hardware		

01 — THE THESIS

You are selling churn absorption

Red Hat's subscription is not access to Linux. It is a promise to absorb upstream churn — kernel changes, CVEs, package breakage — so an enterprise never has to. Customers pay every year because upstream never stops moving.

Your upstream is different and far more volatile: **the user interfaces of the apps a small business runs on**. Toast redesigns the hours screen. Square moves a button. Google Business Profile ships a new layout. Every do-it-yourself automation in America breaks silently, and the owner finds out from an angry customer standing at a locked door.

Yours breaks too. The difference is that it breaks for about six hours.

01 A vendor ships a UI change to their app.	02 Post-action verification fails across the fleet. You know before any customer does.	03 You re-map that one flow. Once.
04 The fix is signed and pushed to every customer overnight.	05 Nobody noticed anything happened.	

That is the entire product and the entire business model in one line. It also explains the economics: the flow library is *shared*. Mapping Toast's hours screen costs you the same whether you have four customers or four hundred. **Your marginal cost for the 401st Toast customer is close to zero, and your value to them is identical.**

It also means this can never be a one-time sale. The churn doesn't stop, so neither does the subscription — and unlike a feature subscription, the customer feels it the week you stop paying attention.

02 — PRODUCT

Four parts, one appliance

Everything below is assembled from open source. Nothing here requires a vendor partnership, an API key, or anyone's permission.

THE HAND A real Android phone Customer's own SIM, customer's own logins, done through each app's normal screens. A physical handset with a real IMEI defeats the emulator detection that blocks every container-based approach —	THE BRAIN The orchestrator box Local models decide what to do and read back what happened. Cloud models are optional and bring-your-own-key. The machine holds the memory, the flow library, and the business's context. Nothing leaves unless the owner sends it.
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and SMS two-factor simply works, because it is a phone.

THE FACE

The wall screen

A ten-foot tile layout everyone already knows from a streaming box. Eight tiles, no menus, no filesystem. Glanceable from across the room, driven by a remote, a phone, or voice.

THE BRAKE

The approval board

Three things need your OK. Each with a draft, a dollar amount, and a screenshot of what the agent is about to do. The agent proposes; the owner disposes. This is the trust mechanism and the support-ticket deflector in one.

03 — STACK

What you actually build on

LAYER	CHOICE	WHY THIS ONE
OS base	Debian stable or CentOS Stream, <code>image mode</code> (bootc / OSTree)	One signed artifact per release. Atomic swap, automatic rollback on failed health check, zero migration scripts, identical on every unit.
Boot & integrity	systemd-boot + unified kernel image, Secure Boot with owner-enrollable keys	Locked by default, unlockable by whoever bought it. Sidesteps GRUB2 entirely.
Update transport	OCI registry + <code>cosign</code> signatures + public transparency log	Pull-only. You publish; devices choose. No pipe into anyone's machine means nothing to compel, steal, or target.
Phone mirror	<code>scrcpy</code> over USB	Already solved. Low-latency Android screen on the Linux box with passthrough input. Do not rebuild this.
Phone control	Custom Android Accessibility Service + ADB transport	Gives you the semantic UI tree — real buttons with real IDs — not pixel guesswork. The sanctioned mechanism for software acting on a user's behalf.
Flow engine	Record → replay → verify, vision model as fallback	Owner performs the task once; you capture the deterministic path. Replay is near-perfect; vision only covers what moved. Gets <code>more</code> reliable with use.
Inference	<code>llama.cpp</code> / vLLM, GGUF weights, LoRA hot-swap	One strong base resident in RAM plus small adapters. Specialization swaps in ~200 ms instead of reloading a 5 GB model.

LAYER	CHOICE	WHY THIS ONE
Silicon	AMD Strix Halo (Ryzen AI Max+), 64–128 GB unified	Only affordable way to give a GPU that much addressable memory — and <code>amdgpu</code> is in-tree, which is what keeps the single-image architecture clean.
Tool protocol	MCP	The emerging standard for agent tool calls. Adopt it rather than inventing a plugin format nobody else speaks.
Human apps	Flatpak, curated remote	Sandboxed, versioned separately from the OS, and the free Linux catalog costs you nothing to include.
Local data	SQLite + <code>sqlite-vec</code> , LUKS full-disk, backup to a destination the customer picks	They get protection from a dead drive; you never hold a key and never hold their books.
Voice & SMS	The SIM in the phone	The business's own number. No third-party telephony vendor, no relay, no per-message cost, no unfamiliar caller ID.

04 — SEQUENCE

Build order

Numbered because it is a real dependency chain, not a menu. Each phase exists to retire one specific risk before you spend money on the next.

P0

Prove remote control on one flow

Weeks 1–4

One laptop, one Android phone, one app, one task. Resist every urge to generalize.

- scrcpy mirroring, ADB transport, accessibility service dumping the UI tree
- Record a human changing business hours in Google Business Profile
- Replay it. Screenshot the end state and read it back to confirm

DONE WHEN You say "update my hours to 9 to 5" and get back a screenshot proving it happened.

P1

Weeks 5-12

Flow library and approval board

Turn one trick into a system. This is where the product becomes real.

- Record-and-replay engine with per-app flow definitions
- Verification on every action — never report success without visual proof
- Approval queue UI, first version, on a plain monitor
- Ten flows across three real businesses in your own city

DONE WHEN Three businesses run five flows each, unattended, for two straight weeks.

P2

Weeks 13-20

Ship software, not hardware

Sell it before you manufacture anything. Customers bring their own mini PC and phone; you charge the subscription. This retires demand risk for the price of a Stripe account instead of the price of a container of inventory.

- Installer, licensing, billing, onboarding that a non-technical owner can finish alone
- Twenty to forty flows across two verticals — pick restaurants and one service trade

DONE WHEN Twenty-five businesses are paying monthly and you have not shipped a single box.

P3

Weeks 21–32

Build the churn machine

The actual company. Everything before this was a demo; this is the thing customers renew for.

- Fleet-wide breakage detection from verification failures
- One-fix-maps-to-all pipeline with signed flow packs
- Weekly release train: canary 5%, staged rollout, human-readable changelog
- A certification rack running every supported app version

DONE WHEN A real app redesign is detected, re-mapped, and pushed fleet-wide inside 24 hours — and no customer opens a ticket.

P4

Weeks 33–48

The appliance

Only now does hardware make sense, because you know exactly what people use it for.

- bootc image pipeline, signing in an HSM, transparency log, staged rollout controller
- Strix Halo box with a powered phone cradle and charge limiting
- Ten-foot tile UI in Wayland kiosk mode
- ODM order, factory provisioning, certification suite in CI

DONE WHEN One hundred units pass the full certification suite and ship to existing software customers first.

P5

Year 2

Widen

- Curated local model catalog — the one-click Hugging Face, quantized and tested on your exact box
- Voice and inbound call handling on the business line
- Regulated tier for clinics and firms: retention controls, compliance logging
- Multi-location fleet SKU where the customer holds the management keys

05 — OPERATIONS

The week that is the subscription

If this loop runs, the business works. If it stops for a month, customers start finding broken flows before you do — and that is the only way this product dies.

MON Fleet health. Which verifications failed over the weekend, and on which app versions.	TUE - WED Re-map broken flows. Test against the certification rack, not a developer's phone.	THU Build and sign the flow pack. Canary to 5% of the fleet. Watch.
FRI Staged rollout to everyone. Changelog written for a restaurant owner, not an engineer.	ONGOING Onboard new apps customers ask for. Add models to the catalog. Grow the library that every customer shares.	

06 — ECONOMICS

What it costs and what it returns

LINE	FIGURE	NOTE
Subscription	\$200 / mo	Flat. Not metered — a decisive advantage over per-token cloud AI for an owner who hates surprise bills.
Software-only tier	\$150 / mo	Customer's own hardware. Your P2 product and your lowest-risk funnel.
Appliance, 64 GB	\$2,495	BOM roughly \$1,100–1,500 at low volume. Hardware is the funnel, not the profit.
Appliance, 128 GB	\$3,495	For businesses running larger local models or several at once.
Phone	\$200–400	Include it. Controlling the handset model is what keeps your flow library testable.
100 customers	\$240K ARR	Plus hardware. Roughly one person's full-time churn loop supports this.
500 customers	\$1.2M ARR	Same flow library. The loop grows sublinearly — this is the leverage.
Replaces	\$800–2,500 / mo	A part-time assistant, or the stack of disconnected subscriptions the owner is already paying for.

The number that matters most is not ARR — it is **flows per customer**. A business running two flows will churn. A business running fifteen has moved its operations onto your box and will not leave. Instrument that from day one and treat it as the only retention metric.

07 — RISK REGISTER

What will actually go wrong

App anti-automation detection

TEST WEEK ONE Some apps — banking especially, via Play Integrity — refuse to run with an active accessibility service. This is cheap to check and it defines your addressable market. Run every target app on a real handset before you write a line of the flow engine, and keep a per-app go / no-go list.

Flow brittleness

REPLAY, THEN VERIFY Pure vision agents land somewhere around 60–85% on novel multi-step tasks — fine for hours, unacceptable for invoices. Recorded deterministic paths plus mandatory end-state verification is what takes it to shippable, and a failed verification is a fleet signal, not a customer complaint.

Three hard products at once

SHIP P2 BEFORE P4 Immutable OS, model orchestrator, and GUI agent are each a company. Only the third one sells. Software on the customer's own hardware buys you revenue and evidence before you commit to inventory you cannot return.

Phone as a wear item

CHARGE LIMITING A handset pinned at 100% forever swells. Cap charge in the cradle, schedule reboots, watch for app auto-updates changing UIs overnight, and treat the phone as a serviceable part with a replacement path.

Overselling local model quality

NEVER CLAIM SMARTER Local models trail frontier cloud models and customers will notice within a week. Sell what is actually true and actually unavailable elsewhere: private, flat-fee, offline-capable, and theirs. You already said you are not trying to beat ChatGPT — keep that in the marketing, not just the strategy.

Single-vendor concentration

TWO VERTICALS EARLY If 80% of your flows live inside one POS vendor, that vendor's roadmap is your roadmap. Spread across at least two unrelated trades by P2 so no single app redesign can take down the business.

08 — START HERE

The next thirty days

- 01** Buy one unlocked mid-range Android handset and put a working SIM in it. Buy one mini PC with 32 GB or more.
 - 02** Get scrcpy mirroring the phone and ADB sending input. One afternoon.
 - 03** Write the accessibility service that dumps the UI tree as structured text. This is the foundation of everything.
 - 04** Pick exactly one flow: business hours in Google Business Profile. Record a human doing it.
 - 05** Replay it. Screenshot the result. Read the screenshot back and confirm the hours changed.
 - 06** Film the whole thing in one unbroken take, thirty seconds, no edits.
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That video is the company

It is your first ten customers, your first hire, and your fundraiser. Someone says "update my hours" out loud and a business's real listing changes on a real account, end to end, with proof. Nobody has to be told what it means.

Build it before you buy a single piece of inventory, register a single trademark, or design a single screen.

Working plan, revision one. Phase durations assume a small team and will move; the dependency order between phases should not. Every component named here is open source and available today.